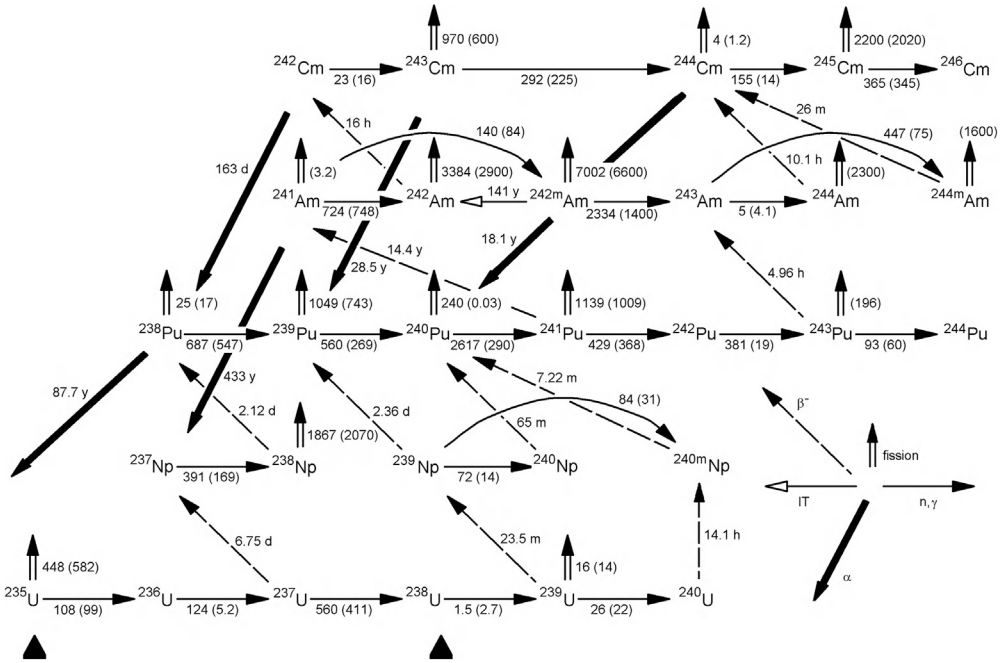


Figure 19.5 Nuclear reactions in irradiation of uranium. Figures along the arrows are half lives or effective cross-sections (b) for a typical power LWR, with thermal (0.025 eV) data within parentheses.

high Σ_a) reduces the effective cross sections below those given in Table 19.2 and Figure 19.5 for isotopes with high concentration in the fuel pins.

19.4. THE FISSION FACTOR

In thermal reactors ^{235}U is consumed mainly by fission and radiative capture

$$^{235}\mathrm{U} + \mathrm{n_{th}} \begin{cases} \xrightarrow[582\ \mathrm{b}]{\sigma_{n,f}} & 2\ \mathrm{FP} + \nu\mathrm{n} \\ \xrightarrow[99\ \mathrm{b}]{\sigma_{n,\gamma}} & ^{236}\mathrm{U} \\ & \downarrow \alpha \end{cases} \tag{19.4}$$

^{236}U decays through α-emission. Due to its long half-life (2.3×10^7 y) it accumulates in the reactor. ^{236}U may capture a neutron, forming ^{237}U, which over a few days time decays to ^{237}Np. The *neutron yield per fission*, ν, is an average value which depends on the neutron energy; it is 2.42 for thermal fission of ^{235}U (see Table 19.2).

If we define the ratio

$$\alpha = \sigma_\gamma/\sigma_f = \Sigma_\gamma/\Sigma_f \tag{19.5}$$

(cf. (11.6)) the probability that the captured neutron gives rise to fission is $\sigma_f/(\sigma_f + \sigma_\gamma) = 1/(1 + \alpha)$. The number of neutrons produced for each neutron captured is

$$\eta = \nu/(1 + \alpha) \tag{19.6}$$

where η is the *neutron yield per absorption*, also called the *fission factor*. A primary requirement for a chain reaction is that $\eta > 1$. Table 19.2 contains η-values for the most important nuclides. At thermal energies η is highest for ^{233}U, while for fast neutrons η is highest for ^{239}Pu.

In mixtures of isotopes the macroscopic cross section Σ must be used in calculating α. Since for natural uranium $\Sigma_f = 0.72 \times 582$ per 100 and $\Sigma_\gamma = (0.72 \times 99 + 99.3 \times 2.70)$ per 100, one obtains $\alpha = 0.81$ and $\eta = 1.34$, which means a chain reaction is possible with thermal neutrons in natural uranium.

19.5. NEUTRON MODERATION

The mode of moderation of the neutrons is one of the crucial design features of a thermal reactor. The fast fission neutrons lose their kinetic energy through elastic scattering with the atoms of the moderator and construction material. In §10.5 the equations are given for the energy change in such collision processes, but only for head-on collisions. Because most collisions involve angular scattering, the number of collisions required to reduce a fast neutron to thermal energy is larger. The *average logarithmic energy decrement* is given by

$$\xi = \overline{\ln\left(E_n/E_n'\right)} = 1 + \left[(A-1)^2/(2A)\right]\ln[(A-1)/(A+1)] \tag{19.7}$$

where E_n' is the neutron energy after collision with a moderator atom of mass number A. From Table 19.3 it is seen that "light" water (i.e. H_2O) is most effective in reducing the

Table 19.3 Physical constants of some moderator materials

Property	H_2O	D_2O	Be	C
$N_0 \times 10^{-30}$ (atoms m^{-3})	0.0333	0.0332	0.123	0.0837
ρ (kg m^{-3})	997	1104	1848	1670
σ_a(th) (barns)	0.66	0.0013	0.0076	0.0035
Σ_a(th) (m^{-1})	2.2	0.0044	0.094	0.029
σ_s(epith) (barns)	49	10.6	5.9	4.7
ξ	0.927	0.510	0.206	0.158
$\xi \times \Sigma_s$(epith)/Σ_a(th)	68	4095	160	211
L_m^2(m^2)	0.000713	1.51	0.043	0.287
τ (m^2)	0.0031	0.0125	0.0097	0.0349